Pixel-Based Crop Classification

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Problem Statement

Most of the food we eat comes from large scale agriculture, so it is naturally very important to understand what exactly is being grown where. Knowing what crops are currently being grown over a wide area allows us to predict and regulate markets to prevent shortages, while also being aware of potentially devastating blights due to overreliance on a specific source. Knowing where each crop is also allows us to prevent and respond quickly to disasters. It is extremely difficult to coordinate every farm in the country enough to get accurate and up to date information, so the USDA has become interested in developing a way to automatically determine what crop a plot of land contains based on satellite imagery. The result of this research is a map covering the whole of the United States at a 10x10 meter resolution describing what (if any) crop is being grown. This is called the cropland data layer (CDL). For our project, we will replicate a portion of the CDL by training a classifier on publicly available data.

Dataset Description

The data we will use to predict will come from the Landsat imagery system. Images are available from EarthExplorer in a variety of formats. These are provided as images with 10m resolution, where each pixel contains information from nine spectral bands. This should be sufficient information to predict the correct label, but we may also try adding information derived from the excavation, such as aspect or slope. This can be sourced from the SRTM dataset (also from EarthExplorer).

The ground truth data may end up being a combination of several datasets. To start, we have data from Washington and California. These datasets both have over ten labels, not all of which line up logically, so some of the initial work will be deciding how to combine them.

The data will be overlaid spatially using ArcGIS Pro, then exported to a tabular dataset where each row represents a pixel. It will have at least ten columns, one of which will be a label.

Proposed Methodology

Many projects using pixel based classification find that the data is well represented by decision tree based models. We will test several such algorithms, including a single tree, a random forest, and a gradient boosted forest.

Before training, we will split some of the data into a test set to be used to evaluate each model. We may need to split the data spatially instead of randomly to prevent bias from the models having trained on pixels adjacent to the test pixels.

In addition to directly testing various decision tree models, we will also explore feature manipulation using PCA and common hyperspectral indices like NDVI. It may also be interesting to look at the results of clustering the data and see if the groups follow logical patterns based on the types of plant.

Depending on our methodology, we expect to evaluate our data using different metrics, analyzing with a combination of silhouette score, Davies-Bouldin Index, and Adjusted Rand Index. Using classification would lead us to evaluating the accuracy and precision of our model when compared against the real solutions.

Expected Challenges + Mitigation

Our goal for this project is to base the model on manually obtained data to replicate the research of the USDA. Ideally we will be able to combine datasets from multiple locations, but it may turn out that the differences in the labels chosen/available do not overlap enough to develop a useful model in a reasonable amount of time. In that case, we will pick a single location and develop a model for the crops grown in that specific area. The dataset published by the Washington State Department of Agriculture will be sufficient for this.

Timeline + Task Distribution

	Alex	Jordan
Oct 24 - Nov 1	Obtain datasets and manipulate into usable format	Initial research into what methods to use
Nov 1 - Nov 15	Implement and test feature augmentations	Begin testing/analyzing ML models against the data
Nov 15 - Dec 1	Analyze results and begin the final report	Analyze results and begin the final report
Dec 1 - Dec 12	Finalize report (will split up sections once available)	Finalize report (will split up sections once available)

References

United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS), 20260227, Cropland Data Layer: USDA NASS, USDA NASS Marketing and Information Services Office, Washington, D.C. Online Links: https://croplandcros.scinet.usda.gov/

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